

1. Ordinary generating function for the number of solutions

We wish to have an ordinary generating function whose coefficient of x^n counts the number of integer solutions to

$$c_1 + c_2 + c_3 + c_4 + c_5 = n,$$

subject to

$$2 \leq c_1 \leq 4, \quad c_2 \text{ is a multiple of } 3, \quad 3 \leq c_i \leq 8 \text{ for } 3 \leq i \leq 5.$$

Similar to the in-class examples, we will build the generating function one variable at a time.

From c_1

Since c_1 can be 2, 3, or 4, its generating function is

$$x^2 + x^3 + x^4.$$

From c_2

Since c_2 is a multiple of 3, we have

$$c_2 \in \{0, 3, 6, 9, \dots\},$$

so its generating function is

$$1 + x^3 + x^6 + x^9 + \dots = \frac{1}{1 - x^3}.$$

From c_3, c_4, c_5

For each of c_3, c_4, c_5 , the allowed values are 3, 4, 5, 6, 7, 8, so each contributes

$$x^3 + x^4 + x^5 + x^6 + x^7 + x^8.$$

Since there are three such variables, together they contribute

$$(x^3 + x^4 + x^5 + x^6 + x^7 + x^8)^3.$$

Final generating function

Therefore, we arrive at our ordinary generating function:

$$F(x) = (x^2 + x^3 + x^4) \frac{1}{1 - x^3} (x^3 + x^4 + x^5 + x^6 + x^7 + x^8)^3.$$

With simple algebra, we have:

$$x^2 + x^3 + x^4 = x^2(1 + x + x^2),$$

and

$$x^3 + x^4 + x^5 + x^6 + x^7 + x^8 = x^3(1 + x + x^2 + x^3 + x^4 + x^5),$$

so

$$F(x) = x^{11}(1 + x + x^2) \frac{(1 + x + x^2 + x^3 + x^4 + x^5)^3}{1 - x^3}.$$

Therefore, the number of solutions to

$$c_1 + c_2 + c_3 + c_4 + c_5 = n$$

is the coefficient

$$[x^n]F(x).$$

In particular, the number of solutions to

$$c_1 + c_2 + c_3 + c_4 + c_5 = 30$$

is

$$\boxed{[x^{30}] (x^2 + x^3 + x^4) \frac{1}{1 - x^3} (x^3 + x^4 + x^5 + x^6 + x^7 + x^8)^3}.$$

2. Sequences and generating functions

2(A)(I) Find the sequence generated by $f(x) = \frac{4x^2}{2 - 6x}$

We first simplify:

$$f(x) = \frac{4x^2}{2 - 6x} = \frac{2x^2}{1 - 3x}.$$

Now use the geometric series formula

$$\frac{1}{1 - 3x} = \sum_{n \geq 0} 3^n x^n.$$

Multiplying by $2x^2$ gives

$$f(x) = 2x^2 \sum_{n \geq 0} 3^n x^n = \sum_{n \geq 0} 2 \cdot 3^n x^{n+2}.$$

We wish to let $m = n + 2$. Then for $m \geq 2$, the coefficient of x^m is

$$2 \cdot 3^{m-2}.$$

Trivially, the coefficients of x^0 and x^1 are 0.

So the generated sequence is defined as:

$$\boxed{(0, 0, 2, 6, 18, 54, 162, \dots)}.$$

Equivalently,

$$a_0 = 0, \quad a_1 = 0, \quad a_n = 2 \cdot 3^{n-2} \quad (n \geq 2).$$

2(A)(II) Find the sequence generated by $g(x) = \frac{2}{1+x^2} - e^{3x}$

We begin by expanding each term separately:

First,

$$\frac{2}{1+x^2} = 2 \sum_{n \geq 0} (-1)^n x^{2n} = 2 - 2x^2 + 2x^4 - 2x^6 + \dots$$

Second,

$$e^{3x} = \sum_{n \geq 0} \frac{(3x)^n}{n!} = \sum_{n \geq 0} \frac{3^n}{n!} x^n.$$

Therefore

$$g(x) = \sum_{n \geq 0} b_n x^n,$$

where

$$b_n = \begin{cases} 2(-1)^{n/2} - \frac{3^n}{n!}, & n \text{ even,} \\ -\frac{3^n}{n!}, & n \text{ odd.} \end{cases}$$

The first several terms are:

$$\begin{aligned} b_0 &= 2 - 1 = 1, \\ b_1 &= -3, \\ b_2 &= -2 - \frac{9}{2} = -\frac{13}{2}, \\ b_3 &= -\frac{27}{6} = -\frac{9}{2}, \\ b_4 &= 2 - \frac{81}{24} = 2 - \frac{27}{8} = -\frac{11}{8}. \end{aligned}$$

So the sequence begins

$$\boxed{\left(1, -3, -\frac{13}{2}, -\frac{9}{2}, -\frac{11}{8}, \dots\right)}.$$

2(B)(I) OGF for the sequence $(2, 0, 2, 0, 2, 0, \dots)$

Let

$$a_n = \begin{cases} 2, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

We observe the ordinary generating function is

$$A(x) = 2 + 2x^2 + 2x^4 + 2x^6 + \dots$$

Simply, we can see this is a geometric series with first term 2 and ratio x^2 , so

$$A(x) = \frac{2}{1-x^2}.$$

Hence the ordinary generating function is

$$\boxed{\frac{2}{1-x^2}}.$$

2(B)(II) OGF for the sequence $(6, 27, 128, 629, \dots) = (1 + 5, 2 + 5^2, 3 + 5^3, 4 + 5^4, \dots)$

From the pattern, the n th term (starting at $n = 1$) is

$$a_n = n + 5^n.$$

So, we find the ordinary generating function to be

$$A(x) = \sum_{n \geq 1} (n + 5^n)x^n = \sum_{n \geq 1} nx^n + \sum_{n \geq 1} (5x)^n.$$

We use the standard formulas (as defined in lecture)

$$\sum_{n \geq 1} nx^n = \frac{x}{(1-x)^2}, \quad \sum_{n \geq 1} (5x)^n = \frac{5x}{1-5x}.$$

Therefore

$$\boxed{A(x) = \frac{x}{(1-x)^2} + \frac{5x}{1-5x}}.$$

3. Coefficient of x^5 in $f(x) = \frac{2}{3x^2 - 4x + 1}$

We factor the denominator:

$$3x^2 - 4x + 1 = (3x - 1)(x - 1) = (1 - 3x)(1 - x).$$

So

$$f(x) = \frac{2}{(1-3x)(1-x)}.$$

Now we will use partial fractions:

$$\frac{2}{(1-3x)(1-x)} = \frac{A}{1-3x} + \frac{B}{1-x}.$$

Then

$$2 = A(1-x) + B(1-3x).$$

Expanding,

$$2 = (A+B) + (-A-3B)x.$$

Matching coefficients gives

$$A + B = 2, \quad -A - 3B = 0.$$

From the second equation,

$$A = -3B.$$

Substitute into the first:

$$-3B + B = 2 \implies -2B = 2 \implies B = -1,$$

so

$$A = 3.$$

Thus

$$f(x) = \frac{3}{1-3x} - \frac{1}{1-x}.$$

Expand each term as a power series:

$$\frac{3}{1-3x} = 3 \sum_{n \geq 0} 3^n x^n = \sum_{n \geq 0} 3^{n+1} x^n,$$

and

$$\frac{1}{1-x} = \sum_{n \geq 0} x^n.$$

Therefore

$$f(x) = \sum_{n \geq 0} (3^{n+1} - 1)x^n.$$

So the coefficient of x^5 is

$$3^6 - 1 = 729 - 1 = 728.$$

Hence

$$\boxed{728.}$$

4. Closed form of the OGF for the convolution of $(n^2)_{n \geq 0}$ and $(n)_{n \geq 0}$

Let

$$a_n = n^2, \quad b_n = n.$$

If c_n is the convolution of these sequences, then we know

$$c_n = \sum_{k=0}^n a_k b_{n-k} = \sum_{k=0}^n k^2 (n-k).$$

The ordinary generating function of the convolution is the product of the ordinary generating functions:

$$C(x) = A(x)B(x),$$

where

$$A(x) = \sum_{n \geq 0} n^2 x^n, \quad B(x) = \sum_{n \geq 0} n x^n.$$

Now we wish to use the standard formulas (as defined in lecture):

$$\sum_{n \geq 0} n x^n = \frac{x}{(1-x)^2}, \quad \sum_{n \geq 0} n^2 x^n = \frac{x(x+1)}{(1-x)^3}.$$

Therefore

$$C(x) = \frac{x(x+1)}{(1-x)^3} \cdot \frac{x}{(1-x)^2} = \frac{x^2(x+1)}{(1-x)^5}.$$

So the closed form is

$$\boxed{\frac{x^2(x+1)}{(1-x)^5}}.$$